# AN UNRAMIFIED FLAT MORPHISM OF SCHEMES IS FORMALLY ETALE: THE CONORMAL SHEAF DIES BY OPEN-DIAGONAL LOCALIZATION

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 1625-1628 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 1630-1652, the complete author proof (21 body lines).

*Editorial note.* Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Counted A side only (spaces twin proof incomplete per ruling v3).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. Morphisms, lemma diagonal-unramified-morphism; chapter lemmas unramified-formally-unramified, base-change-formally-unramified, universal-thickening-fibre-product-flat, universal-thickening-localize (schemes form in-source).

*Source reference.* Original lemma 02HB: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 847–849. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 04F9: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1282–1294. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 04FA: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1310–1319. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 02HE: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 975–979. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 01U9: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1652–1652. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

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*Date*: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original lemma 02GE: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1652–1652. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *An unramified flat morphism is formally étale.*

*Proof.* Say $X \to S$ is unramified and flat. Then $\Delta : X \to X \times_S X$ is an open immersion, see Morphisms, Lemma 02GE. We have to show that $\mathcal{C}_{X/S}$ is zero. Consider the two projections $p, q : X \times_S X \to X$. As $f$ is formally unramified (see Lemma 02HE), $q$ is formally unramified (see Lemma 02HB). As $f$ is flat, $p$ is flat, see Morphisms, Lemma 01U9. Hence $p^*\mathcal{C}_{X/S} = \mathcal{C}_q$ by Lemma 04F9 where $\mathcal{C}_q$ denotes the conormal sheaf of the formally unramified morphism $q : X \times_S X \to X$. But $\Delta(X) \subset X \times_S X$ is an open subscheme which maps isomorphically to $X$ via $q$. Hence by Lemma 04FA we see that $\mathcal{C}_q|_{\Delta(X)} = \mathcal{C}_{X/X} = 0$. In other words, the pullback of $\mathcal{C}_{X/S}$ to $X$ via the identity morphism is zero, i.e., $\mathcal{C}_{X/S} = 0$. □